

Project Title: NLP & Sentiment Analysis

Objective:

The goal of this project is to develop a machine learning model capable of analyzing unstructured text data

(such as product reviews) and classifying it into sentiment categories (Positive or Negative).

Project Description:

This project focuses on Natural Language Processing (NLP) techniques to preprocess textual data and convert

it into a structured numerical format suitable for machine learning algorithms.

Key Steps:

1. Text Preprocessing:

- Tokenization: Breaking text into individual words.
- Stop-word Removal: Removing common words that do not add meaningful information.
- Lemmatization: Converting words into their base or root form.

2. Feature Engineering:

- Use TF-IDF (Term Frequency-Inverse Document Frequency) to convert text into numerical feature vectors.

3. Model Building:

- Train a classification model such as Naive Bayes or Support Vector Machine (SVM).
- Evaluate model performance using accuracy, precision, recall, and F1-score.

4. Prediction:

- The model should classify new unseen reviews as Positive or Negative.

Tools & Technologies:

- Python
- NLTK / SpaCy
- Scikit-learn
- Pandas, NumPy

Outcome:

A fully functional NLP pipeline capable of performing sentiment analysis on real-world text data.